Practical-7 Sliding Window

AIM: Write a Program to implement flow control at data link Layer using SLIDING WINDOW PROTOCOL simulate the flow of frames from one node to another.

Create a Sender Program with following features:-

1. Input window size from the user
2. Input a Text message from the user.
3. Consider 1 character per frame.
4. Created a frame with following fields
5. Send the frames.
6. Wait for the acknowledgement from the Receiver. [Induce delay in the Program].
7. Reader a file called Receiver-Buffer.
8. Check Ack field for the Acknowledgement number.
9. If the Acknowledgement number is a expected Send new set of frames accordingly. [over write the Sender-Buffer file with new frames] [ELSE if Nack is received, resend the frames accordingly.

Create a receiver file with following features.

1. Reader a file called Sender-Buffer.
2. Check the Frame no
3. If the fame no. are as expected.


n = int(input("Enter number of frames: "))
w = int(input("Enter window size: "))
frames = list(range(1, n + 1))
i = 0
while i < n:
window = frames[i:i+w]
print("\nCurrent Window:", window)
for frame in window:
print("Sending frame:", frame)
for frame in window:
print("Acknowledgement received for frame:", frame)
i += w
print("\nAll frames sent successfully")
Enter number of frames: 5
Enter window size: 3
Current Window: [1, 2, 3]
Sending frame: 1
Sending frame: 2
Sending frame: 3
Acknowledgement received for frame: 1
Acknowledgement received for frame: 2
Acknowledgement received for frame: 3
Current Window: [4, 5]
Sending frame: 4

write the appropriate ACK no. in the Receiver Buffer file. Else write NACK no. in the Receiver Buffer file.

Code flow Control

```
n = int(input("Enter number of frames:"))
w = int(input("Enter window size:"))
frames = list(range(1, n+1))
i = 0
while i<n:
    window = frame[i:i+w]
    Print("\n current window:", windows)
    for frame in window:
        Print("sending frame:", frame)
    for frame in window:
        Print("Acknowledgement received for frame
        :", frame)
        i += w
    Print("\n All frames sent successfully")
```

Result :-

Thus Sliding Window protocol code is written and executed in python.